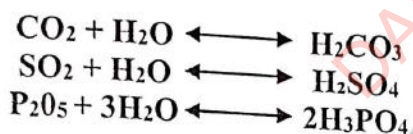


CHAPTER - 8ACIDS, BASE AND SALTS"ACIDIC, BASIC AND AMPHOTERIC SUBSTANCES"

Oxides of metals and non-metals are not themselves acid or base but when dissolved in water, they form acidic or alkaline solution. On the basis of this fact, oxides are classified into three main groups.

- Metal oxides such as Na_2O , K_2O , CaO etc. readily mixed in water to produce alkaline solution.
- Non-metal oxides such as CO_2 , SO_2 , P_2O_5 etc. when mixed in water, they produce acidic solution.



- Certain metal oxides such as ZnO , Al_2O_3 etc. are not water soluble. However, they have ability to react with both acid and base. These are on the borderline of both acid and base and known as **amphoteric substances**.



Beside oxides, certain other substances are also acidic and basic in nature for example Na_2CO_3 gives alkaline solution in water and NH_4Cl gives acidic solution.

"THE BRONSTED-LOWRY THEORY OF ACIDS AND BASES"

In 1923 J.N Bronsted (Danish Chemist) and J.M Lowry (British Chemist) individually propose this general approach to explain acids and bases as an extension of Arrhenius concept.

According to this theory,

"Acid is a species which tends to donate proton (protogenic) whereas base is a species that accept proton (protophillic). Further, an acid base reaction is the transfer of proton from acid to base".

In the light of this definition all hydrogen containing substances (either molecules or ions) which are capable of giving up proton (HCl , HNO_3 , H_2SO_4 , H_2O , H_3O^+ , NH_4^+ etc.) are gathered in the list of acids whereas all those substances which have ability to accept proton (NH_3 , H_2O , OH^- , Cl^- , CO_3^{2-} , HSO_4^-) are categorized as bases.

Proton Donors and Acceptors:

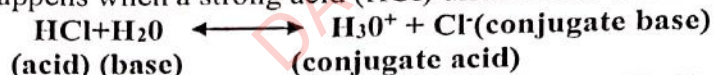
According to Bronsted Lowry concept,

The necessary need for an acid base reaction is that one substance can donate a proton while other substance is able to accept it. In this proton transfer process both acid and base work together.

This means if one species is an acid the other always be a base.

Example #1:

Let us consider what happens when a strong acid (HCl) dissolves in water and undergo ionization.

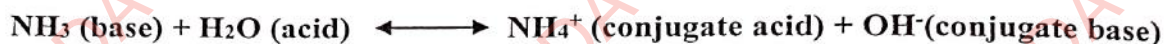


Since HCl donate a proton, it acts as an acid on the other hand water (H₂O) accept a proton and serve as base.

Look at the reverse of this acid base reaction where hydroxonium ion (H₃O⁺) reacts with chloride ion (Cl⁻). In the reverse reaction, H₃O⁺ act as conjugate acid while Cl⁻ ion serves as conjugate base. Thus the equilibrium of the above reaction represents two acids and two bases, one on either side of the reaction arrow.

Example #2:

Now see the ionization of ammonia



While studying above two acid base reactions it is interesting to note that water serves as base when HCl is dissolved into it but acts as acid when ammonia is added.

Relative Strength of Bronsted Acid and Base:

The direction of an acid base reaction depends on the relative strengths of concerned acid and base and the reaction is generally goes towards the direction of weaker acid or base. According to Bronsted-Lowry perception, the strongest acid has the weakest conjugate base and the strongest base has weakest conjugate acid. Look at again the acid base reaction. Since HCl is a strong acid, it donates proton to water molecules completely and forms a conjugate base of HCl which is Cl⁻ and conjugate acid of H₂O which is H₃O⁺.

In term of relative strength of two acids (HCl and H₃O⁺), HCl is stronger than H₃O⁺ because it lose proton more readily than H₃O⁺.

Limitations of Bronsted-Lowry Theory:

- This theory cannot explain the reaction between acidic oxides (CO₂, SO₂, SO₃) and the basic oxides like (CaO, BaO, MgO) in which no involvement of proton transfer.
- Certain substances like AlCl₃, BF₃ etc. do not involve in the proton transfer but they function as acid.

"DEGREE OF IONIZATION"**DEFINITION:**

"It is the ratio of no. of dissociation particles and the total no. of particles."

$$\alpha = \frac{\text{No. of dissociated particle}}{\text{Total no. of particle}}$$

Strong electrolyte (HCl, NaOH, NaCl) have high degree of ionization and weak electrolyte have low degree of ionization. The conductance depends upon degree of ionization of electrolyte. Greater is the degree of ionization greater the ionization.

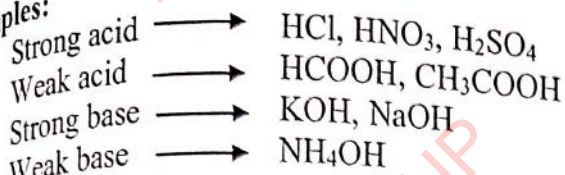
"STRENGTH OF ACID AND BASE"**DEFINITION:**

"The capacity of all acids and base to ionize into H⁺ and OH⁻ ions in an equimolar solution of water"

Strong Acid and Strong Bases:

Acids and bases which fully ionize in aqueous solution are called strong acid and strong bases. On the other hand those acids and bases which ionize to minimum extent are called weak acids and weak bases.

Examples:



HCl and H₂SO₄ are completely dissociated indicating that they are strong acids, the word strength refers to proton donating ability of an acid or its degree of ionization is expressed in terms of degree of dissociation or percentage dissociation.

$$\alpha = \frac{\text{No. of dissociated particle}}{\text{Total no. of particle}}$$

The acids and bases having high percentage of dissociation are strong and those with low percentage dissociation are a weak

"IONIZATION CONSTANT OF WATER (K_w)"

Though water is chemically non electrolytic substance, however it conducts electricity to a very small extent due to its self-ionization ability. During the reaction proton from one water molecule is transferred to another water molecule as given in the following reaction.



equilibrium constant may be written as,

$$\frac{[\text{H}_3\text{O}^+][\text{OH}^-]}{[\text{H}_2\text{O}][\text{H}_2\text{O}]}$$

$$[\text{H}_2\text{O}]^2 = [\text{H}_3\text{O}^+][\text{OH}^-] \quad (\text{Conc. Of Water is constant})$$

$$= [\text{H}_3\text{O}^+][\text{OH}^-]$$

At 25°C, the value of K_w is 1.0 x 10⁻¹⁴. This very small value of K_w shows that self-ionization water occurs to a very small extent. Since the number of [H⁺] and [OH⁻] in pure water are equal, we determine the concentration of these ions by using above equation of K_w. Thus in pure water the concentration of each [H⁺] and [OH⁻] at 25°C is 1 x 10⁻⁷ M.

The value of K_w varies with temperature and a rise of temperature increases the value of K_w."

"pH, pOH and pK_w "

The measurement of the concentration of aqueous H^+ and OH^- ions is a task to estimate the strength of acids and bases because the conc. of H^+ and OH^- ions are very low in aqueous solution specifically when a weak acid or base is dissolved. To cope up this difficulty Sorenson (1909) introduced a logarithmic scale since logarithm is very useful way in representing very small and very large quantities.

"pH is the negative logarithm of molar concentration of H^+ ions and pOH is the negative logarithm of molar concentration of OH^- ions in aqueous solution at given temperature."

Since pH is a logarithmic value, it has no unit.

pH scale represents numbers between zero (very acidic) to 14 (very alkaline) and specify the acidity or basicity of aqueous solution. A solution having $pH = 7$, represents the point of neutrality and it evident that solution contains equal conc. of H^+ ions and OH^- ions. The behavior of different solutions with respect to pH concept is summarized below.

Although, the pH of distilled water is 7 at $25^\circ C$, however it increases with the rise of temperature since temperature increases the extent of ionization.

The dissociation constant of water is represented as,

$$K_w = [H^+][OH^-]$$

By taking negative logarithm on both sides

$$(-\log K_w) = (-\log [H^+]) + (-\log [OH^-])$$

Since K_w at $25^\circ C$ is 1×10^{-14} .

Therefore,

$$pH + pOH = 14$$

MEASUREMENT OF pH:

Following are the methods to measure the pH of a solution

i. Using universal indicator:

pH of a solution is determined by adding few drops of universal indicator. Universal indicator has been obtained by mixing various indicators. This indicator indicates different colors of solution at different acidities. Universal indicator related to pH values are as follows.

Red	orange	yellow	green	blue	purple
0	3	6	7	8	14

ii. Using pH meter:

A pH meter has a glass electrode which when dipped into solution, will give direct reading of the pH value on the meter.

1.0	1.2	2.9	6.6	7.0	7.3	7.9-8.4	10.5	11.6	13
HCl	H_2SO_4	CH_3COOH	Milk	Distilled water	Blood	Sea Water	MEX of Magnesia	Na_2CO_3	NaOH
← Acidic			Neutral		Basic →				

"LEVELING EFFECT"

"Strong acids like HCl, HBr, HI etc. when dissolve in water, they all exhibit apparently equal strength due to the fact that water play basic medium to get them ionized completely hence water serves as leveling solvent for these acids and the phenomena is called leveling effect."

Therefore, any acid stronger than hydroxonium ion (H_3O^+) completely reacts with water to form H_3O^+ ion and the corresponding conjugate base. This can be attributed to the fact that H_3O^+ is the strongest acid that can possibly exist in an aqueous solution.

For example HCl completely react with water to form H_3O^+ and Cl^- .



The similar concept goes for bases as well. Any base stronger than hydroxide ion (OH^-) completely react with water to form OH^- ion and respective conjugate acid. The reason is similar to our former understanding that OH^- ion is strongest base that can possibly exist in any aqueous solution.

For example when sodium amide (NaNH_2) is placed in water then amide ion (NH_2^-) reacts completely with H_2O .

"LEWIS DEFINITION OF ACIDS AND BASES"

According to G.N Lewis (1923)

"Acid is a substance that accepts an electron pair whereas base is a substance that can donates electron pair".

Lewis acid and base is commonly known as electrophillic (electron loving) and nucleophillic (nucleus loving) reagents since they have capability to accept and donate electron pair respectively.

EXPLANATION:

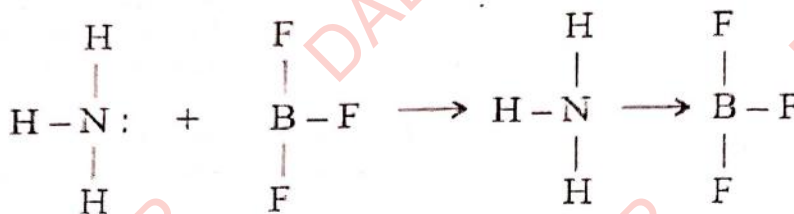
Considering Lewis concept, hydrogen ion (H^+) is an acid because it accepts electron pair. Similarly hydroxide ion (OH^-) is base since it donate lone pair of electron.

Lewis theory does not explain about the transfer of proton from one specie to another however it indicates sharing of electron pair between a donor and acceptor reagent. Generally species (compounds or cations) having less than a full octet of electrons behave like Lewis acids and all species (compounds or anions), having lone pair of electrons behave like Lewis base. The product formed in a lewis acid base reaction possesses a coordinate covalent bond between lewis acid and base and known as adduct.



Several neutral molecules such as AlCl_3 , BF_3 , FeCl_3 etc. serves as Lewis acids due to electron deficiency on the central atom (fewer than eight valence electrons). On the other hand ammonia (NH_3),

Phosphine (PH_3), Water (H_2O) etc. although have their complete octet but possess lone pair of electron that is available to donate hence they serve as Lewis base.



"BUFFERS"

DEFINITION:

"The solution which tends to maintain its pH when acid or alkali is added to it is called buffer solution."

A buffer solution usually consists of mixture of weak acid (weakly dissociating acid) and the salt of acid with strong base or a weak base and its salt with strong acid.

EXAMPLE:

$\text{NaHCO}_3 / \text{H}_2\text{CO}_3$, $\text{NaH}_2\text{PO}_4 / \text{H}_3\text{PO}_4$, and $\text{NH}_4\text{Cl} / \text{NH}_4\text{OH}$, etc.

BUFFER ACTION:

It is well known that at pH of a buffer solution remains almost the same on adding small amount of acid or base.

Addition of Acid:

Suppose HCL is added to a buffer solution of $\text{NaHCO}_3 / \text{H}_2\text{CO}_3$. The following reaction will take place.



Neutral salt (NaCl) and weak acid (H_2CO_3) are formed and only a little change in pH takes place.

Addition of Base:

In the same way if small amount of NaOH is added to this buffer solution it reacts with H_2CO_3 as follows.



Since NaHCO_3 is a much weaker base than NaOH and rise in pH will be very small.

Buffer Capacity:

It is important to realize that buffer solution cannot keep the pH constant if enough acid or base is added into it. The quantity of substance (acid or base) that can be mixed into buffer solution before its pH change is known as buffer capacity.

APPLICATION OF BUFFERS:

Buffer is an important part of many industrial processes such as electroplating, photographic materials dyes and manufacturing of leather. In Bacteriological research culture media are generally buffered to maintain a constant value of pH required for the growth being studies. Buffers are used extensively in analytical chemistry. Buffers are used in calibration of pH meter. pH of human blood is buffered to 7.4 by adding bicarbonates, phosphates and complex protein system.

"SALTS THEIR TYPES AND APPLICATIONS"

"Salt is a substance produce by the reaction of acid and base. It consists of positive ion of base and negative ion of acid".

Salts may be neutral, acidic or basic, depending upon the number and types of ions present.

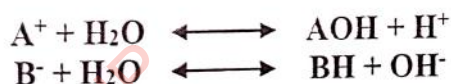
Types of salts formed by neutralization reaction			
Acid	Base	Types of Salts	Neutralization Reaction
Strong	Strong	Neutral	$\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
Strong	Weak	Acidic	$\text{HCl} + \text{NH}_4\text{OH} \rightarrow \text{NH}_4\text{Cl} + \text{H}_2\text{O}$
Weak	Strong	Basic	$\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$
Weak	Weak	Neutral	$\text{CH}_3\text{COOH} + \text{NH}_4\text{OH} \rightarrow \text{CH}_3\text{COONH}_4 + \text{H}_2\text{O}$

"HYDROLYSIS"**DEFINITION:**

"The reaction of cation or anion with water so as to change its pH is known as hydrolysis."

EXPLANATION:

Hydrolysis is a chemical interaction between the ions of electrolyte and water molecules during which the pH of solution becomes changed. In short, hydrolysis occurs when a change in pH takes place.



EXAMPLE:

- i. The aqueous solution of NH_4Cl forms strong acid and weak base.
- $$\text{NH}_4\text{Cl} + \text{H}_2\text{O} \rightarrow \text{NH}_4\text{OH} + \text{HCl}$$

Thus the solution contains higher concentration of H^+ ions which change the pH of water towards acidic which turns blue litmus to red.

- ii. The aqueous solution of Na_2CO_3 forms strong base and weak acid
- $$\text{Na}_2\text{CO}_3 + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2\text{CO}_3$$

Thus the solution contains higher concentration of OH^- ions which changes the pH of water towards basic.

- iii. The aqueous solution of NaCl is neutral as it contains strong acid and strong base and pH of water remain unchanged.

SALT AND TYPES OF SALT:

After getting the concept of hydrolysis we shift towards four possible types of salts.

Type-1: (Salt of Strong acid and Strong base)

These salts are formed by neutralization of strong acid and strong base. Since both cation and anion comes from strong base and strong acid respectively, they do not undergo hydrolysis and hence the pH of their aqueous solution remains 7.

For example: NaCl , K_2SO_4 , NaNO_3 etc.

Type-2: (Salt of Strong acid and Weak base)

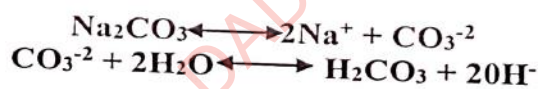
These salts are formed by neutralization of strong acid and weak base. The an ion of these salts comes from strong acid and hence does not react with water, however cat ion which comes from weak base interact with water to give acid solution having pH less than 7. For example: When NH_4Cl dissolves in water, it gives acidic solution.



Other examples of this type of salts are $(\text{NH}_4)_2\text{SO}_4$, CuCl_2 , etc.

Type-3: (Salt of weak acid and strong base)

These salts are formed by neutralization of weak acid and strong base. The cat ion which comes from strong base does not react further with water. The anion which comes from weak acid reacts with water and undergo hydrolysis to give alkaline solution having pH more than 7. For example sodium carbonate when dissolve in water it gives basic solution.



Other examples of this type of salt are CH_3COONa , NaHCO_3 etc.

Definition: (Salt of Weak acid and weak base):

These salts are formed by neutralization of weak acid and weak base. Cation and anion of these salts come from weak base and weak acid respectively therefore they both have ability to undergo hydrolysis in aqueous solution. pH of aqueous solution of these salts may either be more than 7, less than 7 or exactly 7 depending upon the relative extent of hydrolysis of cation and anion.

Example: NH_4CN , $\text{CH}_3\text{COONH}_4$

Application of some salts:

Salts play significant role in the synthesis of a variety of industrial products as well as many of them are used in our daily life. Some applications of salts are mentioned in the following table.

Applications of some typical salts			
Chemical Name	Common Name	Formula	Common Applications
Table salt	Sodium chloride	NaCl	Essential part of Human diet
Epsom salt	Magnesium sulphate	MgSO_4	Use to produce laxative effect to treat constipation
Gypsum salt	Calcium sulphate	CaSO_4	In making plaster of paris
Baking soda	Sodium bicarbonate	NaHCO_3	Use as antacid to relieve heart burn and acid indigestion
Washing soda	Sodium carbonate	Na_2CO_3	Use to remove rigid stains from Laundry
Sal ammoniac	Ammonium chloride	NH_4Cl	Use as expectorant in cough syrup
Muriate of potash	Potassium chloride	KCl	Use in making fertilizers

THEORY QUESTIONS

SHORT QUESTIONS:

1. Define pH & pOH of a solution? Also show that $\text{pH} + \text{pOH} = 14$.
2. Why the aqueous solution of NH_4Cl is acidic and Na_2CO_3 is alkaline.
3. Write down conjugate base of each of the following acids. H_2SO_4 , H_2S , NH_4^+ , HCOOH
4. What is meant by self-ionization of water? Write the expression of K_w . What is its value at 25°C ?
6. Write equation and indicate the conjugate acid-base pairs for the following:
(i) Acetic acid & ammonia (ii) Hydrochloric acid & water

DESCRIPTIVE QUESTIONS:

1. Explain Bronsted-Lowry theory of acids and bases. What is meant by conjugate acid base pair give examples?
2. Define the process of hydrolysis. Explain the behavior of each of the following salts in aqueous solution. (i) K_2CO_3 (ii) $(\text{NH}_4)_2\text{SO}_4$ (iii) NaNO_3
3. What is Buffer solution? Explain how it resists the change of pH by adding small amount of acid and base. Give the applications of buffer solution.
4. Describe Lewis theory of acids and bases. What are the advantages of this theory over Lowry Bronsted theory?
5. What is salt? Explain Acidic, Basic and Neutral salts?

NUMERICALS FROM PAST PAPER

pH / pOH:

1. pH of a solution is 8.4. Calculate the H^+ ion and OH^- ion concentration.
2. Find the pH of 0.0004 mol/dm^3 of HCl (fully dissociated) at 25°C . $\text{pH} = 3.39$
3. What is H^+ and OH^- ion concentration of a solution having pH equal to 7.86?
 $[\text{H}^+] = 1.31 \times 10^{-8} \text{ M}$, $[\text{OH}^-] = 7.24 \times 10^{-7} \text{ M}$
4. Calculate pH of $4.6 \times 10^{-3} \text{ M}$ solution of HCl .
5. The pH of HCl solution is 2.0; find the hydrogen ion concentration of the solution, assuming 100% ionization. $[\text{H}^+] = 0.01 \text{ M}$
6. Calculate H^+ ion concentration of a solution whose $\text{pH} = 5.6$
7. Calculate pH value of 0.004 M NaOH solution.